

# CAPSTONE DEMO TEMPLATE

## Official Capstone Submission & Evaluation Document — PickGuard AI

### 1. Team Name & Problem Formulation

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Team Name:                | PickGuard AI (Author / Engineer: Ruban Raj)                                                                                                                                                                                                                                                                                                                                                                                   |
| Who is the User?          | Fulfilment Centre <b>Pick Operators</b> encountering physical exceptions (missing items, damaged barcodes, bin discrepancies) during active pick waves, along with <b>Warehouse Supervisors / Operations Leads</b> managing escalation queues.                                                                                                                                                                                |
| What is the Bounded Task? | Provide an <b>evidence-grounded, next-best-action decision recommendation</b> in response to natural-language pick exceptions. The agent retrieves verified bin/task/inventory facts and SOP procedures, identifies conflicts, reasons over facts vs inferences, enforces safety boundaries (mutations are blocked), and pauses at a real <b>LangGraph interrupt() checkpoint</b> for human authorization on high-risk cases. |

### 2. Agent Design Architecture

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Graph State:         | PickExceptionState (TypedDict in backend/app/graph/state.py): Tracks query, exception_type, secondary_exception_types, task_id, item_id, location_id, operational_data, sop_evidence, historical_evidence, evidence_package, reasoning, root_cause, next_best_action, action_type, action_status, risk_level, requires_human_review, human_review_payload, review_attempts, audit_log, and errors.                                                                                                                                    |
| Nodes (13 Nodes):    | <b>1.</b> parse_operator_input <b>2.</b> classify_exception <b>3.</b> retrieve_operational_data <b>4.</b> retrieve_sop_evidence <b>5.</b> retrieve_historical_evidence <b>6.</b> build_evidence_package <b>7.</b> llm_reasoning_node <b>8.</b> evidence_fusion_node <b>9.</b> evaluate_safety_policy <b>10.</b> human_review_gate (with interrupt()) <b>11.</b> collect_additional_evidence <b>12.</b> apply_final_decision <b>13.</b> END.                                                                                           |
| Deterministic Tools: | <ul style="list-style-type: none"><li>• get_inventory(item_id, location_id): Database units &amp; status.</li><li>• get_pick_task(task_id): Expected pick quantity &amp; target bin.</li><li>• get_location(location_id): Storage zone &amp; adjacent neighbouring bins.</li><li>• search_similar_incidents(exception_type): Past resolution precedence.</li><li>• escalate_to_lead(task_id, reason): WMS supervisor queue escalation.</li><li>• retrieve_sop_chunks(query, exception_type): ChromaDB vector RAG retrieval.</li></ul> |
| Provider Route:      | Multi-Tier Resilient Routing Hierarchy:<br><b>Tier 1: GroqProvider</b> (Llama-3.3-70B cloud inference for ultra-fast reasoning)<br><b>Tier 2: OllamaProvider</b> (Local offline inference with Llama3 model)<br><b>Tier 3: MimicProvider</b> (Deterministic offline fallback guaranteeing 100% demo uptime without external API keys).                                                                                                                                                                                                |
| Human-Review Point:  | Implemented via real from langgraph.types import interrupt, Command in the human_review_gate node. The graph pauses if requires_human_review == True (triggered by hazardous mutating actions or evidence conflicts). State is saved with MemorySaver checkpointer. Resumption occurs via Command(resume={'decision': 'APPROVE'   'REJECT'   'REQUEST_MORE_EVIDENCE'   'ESCALATE'}).                                                                                                                                                  |

### 3. Demonstration Script (3 Core Cases)

| Scenario                                 | Operator Query & Context                                                                                                                                                                               | System Execution & UI Output                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Normal Case<br>(Low Risk)                | <b>Query:</b> <i>'The item X123 is missing from A15-B04. The system says there are 3 units.'</i><br><b>Task:</b> TASK-1001   <b>SKU:</b> X123   <b>Bin:</b> A15-B04                                    | <ul style="list-style-type: none"><li>• <b>Exception:</b> MISSING_ITEM   <b>Risk:</b> LOW RISK (Green)</li><li>• <b>Evidence:</b> Retrieves bin A15-B04 + neighbours [A15-B03, A15-B05], SOP-MISSING-001.</li><li>• <b>Next-Best Action:</b> CHECK_NEIGHBOURING_LOCATION (Status: RECOMMENDED)</li><li>• <b>HITL Interrupt:</b> None required. Workflow completes automatically.</li></ul>                                                                                                                 |
| Edge Case<br>(Multi-Signal)              | <b>Query:</b> <i>'The item X124 is missing at A12-B03 and the barcode also won't scan.'</i><br><b>Task:</b> TASK-1002   <b>SKU:</b> X124   <b>Bin:</b> A12-B03                                         | <ul style="list-style-type: none"><li>• <b>Exception:</b> Primary MISSING_ITEM + Secondary BARCODE_FAILURE</li><li>• <b>Evidence:</b> Retrieves both SOP-MISSING-001 and SOP-BARCODE-001.</li><li>• <b>Next-Best Action:</b> Recommends manual barcode entry followed by physical sweep.</li><li>• <b>HITL Interrupt:</b> None required. Demonstrates compound signal reasoning.</li></ul>                                                                                                                 |
| Failure / Risky Case<br>(HITL Interrupt) | <b>Query:</b> <i>'TASK-1003 quantity mismatch: System says 10 units of X125 at A20-B02 but I counted 6. Update inventory to 6.'</i><br><b>Task:</b> TASK-1003   <b>SKU:</b> X125   <b>Bin:</b> A20-B02 | <ul style="list-style-type: none"><li>• <b>Exception:</b> QUANTITY_MISMATCH   <b>Risk:</b> HIGH RISK (Red)</li><li>• <b>Action Boundary:</b> ADJUST_QUANTITY is automatically <b>BLOCKED</b>.</li><li>• <b>LangGraph Interrupt:</b> Graph pauses at human_review_gate node.</li><li>• <b>Supervisor Review:</b> Modal presents conflict (10 vs 6). Supervisor selects <b>REJECT</b>.</li><li>• <b>Resume:</b> Graph resumes; state set to REJECTED_BY_HUMAN; inventory remains <b>UNCHANGED</b>.</li></ul> |

### 4. Evaluation Summary Matrix

| Test                | Expected                                                                        | Actual                                                                        | Pass? | Improvement                                                        |
|---------------------|---------------------------------------------------------------------------------|-------------------------------------------------------------------------------|-------|--------------------------------------------------------------------|
| Normal              | Safe suggestion (CHECK_NEIGHBOURING_LOCATION), LOW risk, no interrupt.          | Correct category, LOW risk, RECOMMENDED status, completed automatically.      | PASS  | Add direct map visualization for adjacent bin coordinates.         |
| Edge                | Multi-signal detection (MISSING_ITEM + BARCODE_FAILURE), compound verification. | Detected primary and secondary types; retrieved both SOP procedures.          | PASS  | Add optical barcode image preview recognition in future.           |
| Failure / High-Risk | Action BLOCKED, HIGH risk, LangGraph interrupt() pauses graph.                  | Action BLOCKED, paused at checkpoint, resumed cleanly upon supervisor review. | PASS  | Add dual-supervisor multi-signature approval for high-value items. |

## 5. Reflection: What would you improve with one more week?

- 1. Multi-Modal Vision Integration:** Ingest barcode scanner images and camera feeds directly into the graph to automatically detect packaging damage and barcode label blemishes using multi-modal LLMs.
- 2. Real-Time WMS Webhook Streaming:** Transition from polling REST endpoints to WebSocket/Server-Sent Events (SSE) for instant supervisor push notifications when an interrupt checkpoint triggers.
- 3. Multi-Tier Supervisor Escalation Matrix:** Introduce tiered authorization rules based on unit monetary value (e.g. items > \$500 require Area Manager dual sign-off).
- 4. Reinforcement Learning from Human Feedback (RLHF):** Fine-tune the local reasoning model on historical supervisor approval/rejection outcomes to improve first-pass recommendation accuracy.